Runbook: Rotating the Database TLS Certificates

Runbook ID	RB-112
Revision	4 (approved 2026-09-30)
Owner	Platform Operations, database team
Applies to	pgcluster-prod-a, pgcluster-prod-b and pgcluster-stage
Estimated duration	45 minutes per cluster, no downtime expected

1. Purpose

This runbook describes how to replace the server certificates used by the production PostgreSQL clusters before they expire. Certificates are issued by the internal certificate authority and are valid for 90 days.

2. Prerequisites

- Membership of the db-operators group in the access portal.
- A workstation with the cluster CLI version 2.8 or later.
- An approved change request in the change calendar.
- Confirmation that no backup job is running on the target cluster.

3. Environments

Cluster	Region	Primary host	Replicas	Certificate expiry
pgcluster-prod-a	north-1	db-a-01.int.cwhn.example	2	2026-11-14
pgcluster-prod-b	north-2	db-b-01.int.cwhn.example	2	2026-11-21
pgcluster-stage	north-1	db-s-01.int.cwhn.example	1	2026-12-02

4. Procedure

4.1 Check the current certificate

Run the following command from your workstation and note the expiry date shown in the output.

```
openssl s_client -connect db-a-01.int.cwhn.example:5432 \
    -starttls postgres </dev/null 2>/dev/null \
  | openssl x509 -noout -subject -enddate
```

4.2 Request a new certificate

Create a certificate signing request with the configuration file below, and keep the private key on the database host only.

```
[ req ]
default_bits       = 4096
prompt             = no
distinguished_name = dn
req_extensions     = san

[ dn ]
CN = db-a-01.int.cwhn.example
O  = Calderwood Regional Health Network

[ san ]
subjectAltName = DNS:db-a-01.int.cwhn.example, DNS:pgcluster-prod-a.int.cwhn.example
```

4.3 Install the certificate

1. Copy the signed certificate to /etc/postgresql/tls/server.crt on each replica first.
2. Set the file owner to postgres and the mode to 0600.
3. Reload the configuration on each replica and confirm that replication is still streaming.
4. Fail over to a replica, then repeat the steps on the former primary.

The cluster manager reads its TLS settings from the following file:

```
tls:
  enabled: true
  cert_file: /etc/postgresql/tls/server.crt
  key_file: /etc/postgresql/tls/server.key
  min_protocol: TLSv1.2
reload:
  strategy: rolling
  pause_between_nodes: 60s
```

4.4 Verify

Repeat the check from step 4.1 and confirm that the new expiry date is about 90 days in the future.

```
cwctl cluster status pgcluster-prod-a --show-tls
```

5. Rollback

If clients report handshake failures, restore the previous certificate from the backup directory and reload the configuration.

Symptom	Likely cause	Action
SSL error: certificate verify failed	Intermediate certificate missing from the chain	Append the intermediate certificate to server.crt and reload
Replication lag above 30 s	Replica reloaded during heavy write load	Wait for the lag to recover before moving to the next node
Connection refused on port 5432	Key file permissions too open	Set mode 0600 and restart the service

6. Escalation

Escalate to the on-call database lead if the rotation cannot be completed within the change window.

On-call rota ops-db rotation in the paging tool

Chat channel #db-operations

Change manager Service desk, extension 4410

7. Revision history

Revision	Date	Author	Change
1	2025-03-11	A. Petrescu	First version
2	2025-08-02	M. Oyelaran	Added the staging cluster
3	2026-02-19	A. Petrescu	Replaced the manual reload with the rolling strategy
4	2026-09-30	J. Haddad	Updated the CLI commands for version 2.8